

15. Browser Environment aur Specs

Host Environment

JS alag alag jagah chalti hai (browser, Node.js server, etc.). Har jagah ko **host environment** kehte hain, jo language ke core ke alawa apne extra objects/functions deta hai. Browser page control karne ke tools deta hai, Node.js server-side features.

`window` (root object)

Browser me iske do kaam:

1. JS ka **global object** (global functions = `window` ke methods).
2. **Browser window** ka representation.

```
js  
  
function sayHi() { alert("Hello"); }  
window.sayHi();  
  
alert(window.innerHeight); // window ki height
```

DOM (Document Object Model)

- Page ka poora content **objects** ke roop me. Inhe modify kar sakte ho.
- `document` object entry point hai.

```
js  
  
document.body.style.background = "red";
```

- DOM sirf browser ke liye nahi, server-side tools bhi use karte hain.

CSSOM

CSS rules/stylesheets ko objects ke roop me dikhata hai. Zyaadatar hum sirf **CSS classes** add/remove karte hain, isliye CSSOM kam kaam aata hai.

BOM (Browser Object Model)

Document ke alawa browser ke baaki objects:

- `navigator`: browser aur OS ki info (`navigator.userAgent`, `navigator.platform`)
- `location`: current URL padhna aur redirect karna

```
js
```

```
alert(location.href);  
location.href = "https://wikipedia.org";
```

- `alert / confirm / prompt` bhi BOM ka hissa hain.

Specifications

Spec	Kya describe karta hai
DOM spec (dom.spec.whatwg.org)	Document structure, manipulation, events
CSSOM spec	Stylesheets aur style rules
HTML spec (html.spec.whatwg.org)	HTML language + BOM (<code>setTimeout</code> , <code>alert</code> , <code>location</code>) etc.)

Kuch chahiye to search karo: "**WHATWG** [term]" ya "**MDN** [term]".

Yaad rakho

Browser me JS = **window** (root) -> **DOM** (`document`) + **BOM** (`navigator`, `location`, ...) + **CSSOM**. Ab hum DOM seekhenge kyunki UI me document sabse important hai.